# Lab 5: Quantum Computing Workshop

## 1 Superpostion

### 1.1 Calculations

We start with defining a basis state of $|0\rangle$:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

To put the state in superpostion, the Hadamard gate needs to be applied.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{Hadamard} H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[ \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = |+\rangle \tag{1}$$

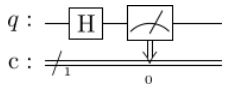
Comparting this to the general form of a qubit state in the given basis (where $|1\rangle = \binom{0}{1}$):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{2}$$

we can say that $\alpha = \beta = \frac{1}{\sqrt{2}}$ which means that $|\alpha|^2 = |\beta|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = .5$, or there is a 50/50 chance of being in ether state.

### 1.2 Code

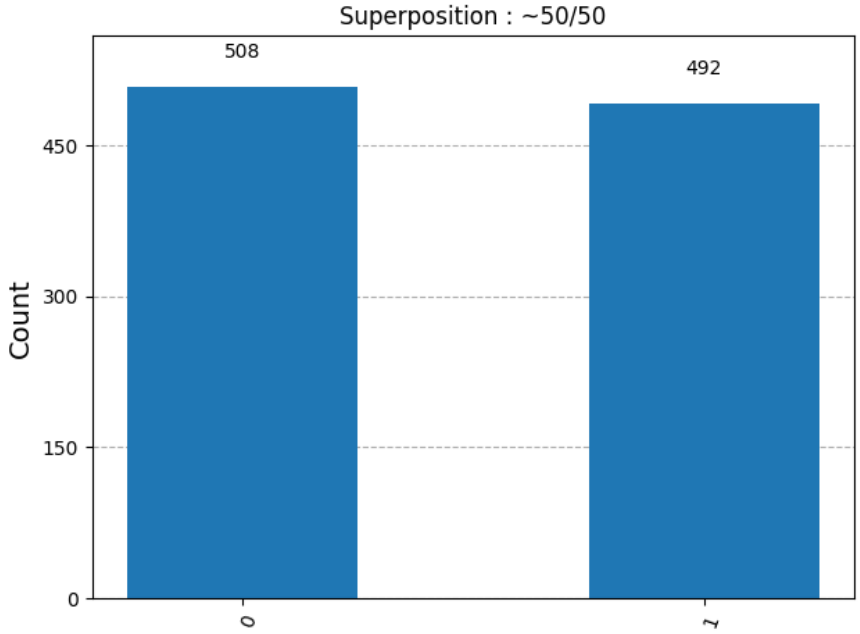
The circuit needed to implement the superpostion is shown below.




For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_super = QuantumCircuit(1,1)
qc_super.h(0)
qc_super.measure(0,0)
shots_super = 1024
job_super = simulator.run([qc_super], shots = shots_super)
plot_histogram(job_super.result().get_counts(0))
```

### 1.3 Results




The result of the simulation shows that the number of shots that were measured in the $|0\rangle$ state was 508 compared to the 492 shots measured in $|1\rangle$. The total number of shots ran was 1000, which means we had $\frac{508}{1000} \approx 51\%$ of the measurements result in $|0\rangle$ and $\frac{492}{1000} \approx 49\%$ of the measurements result in $|1\rangle$ which is within the 5% error margin expected.

## 2 Bell State

### 2.1 Calculations

We now have two qubits which means a new set of basis states are defined.

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

By starting with an initial state of $|00\rangle$ we can apply $H \otimes I$ then CNOT to create a Bell State.

$$(H \otimes I)\,|00\rangle = H\,|0\rangle \otimes I\,|0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right), \tag{3}$$

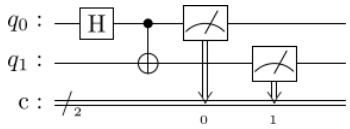
where the result of Eq. 1 was used to calculate $H\,|0\rangle$ and the main property of the Identity matrix was used to calculate $I\,|0\rangle$.

$$\text{CNOT}\,(H \otimes I)\,|00\rangle = \text{CNOT}\frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right) = \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\right)\left(|00\rangle + |10\rangle\right) = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \tag{4}$$

This result is similar to Part 1, but now we should have a 50/50 chance of our result being in $|00\rangle$ or $|11\rangle$ instead of $|0\rangle$ or $|1\rangle$.

## 2.2 Code

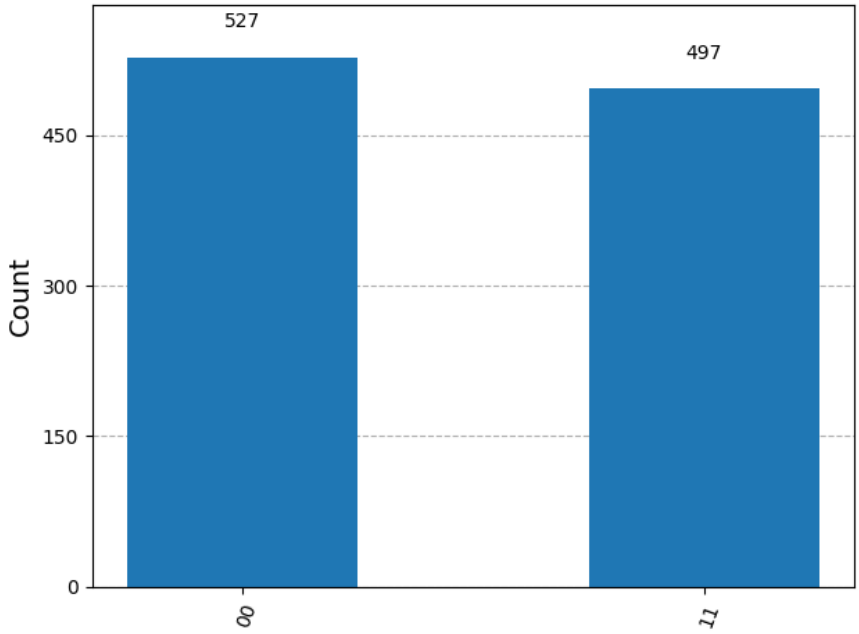
The circuit to implement the bell states is shown below.




For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_bell = QuantumCircuit(2,2)
qc_bell.h(0)
qc_bell.cx(0,1)
qc_bell.measure(0,0)
qc_bell.measure(1,1)
shots_bell = 1024
job_bell = simulator.run([qc_bell], shots = shots_bell)
plot_histogram(job_bell.result().get_counts(0))
```

## 2.3 Results




The result of the simulation shows that the number of shots that were measured in the $|00\rangle$ state was 527 compared to the 497 shots measured in $|11\rangle$. The total number of shots ran was 1024, which means we had $\frac{527}{1024} \approx 51\%$ of the measurements result in $|00\rangle$ and $\frac{497}{1024} \approx 49\%$ of the measurements result in $|11\rangle$ which is within the 5% error margin expected. These percentages add up to 100%, which is expected as the only two results shown in the histogram are $|00\rangle$ and $|11\rangle$.

# 3 Modified Bell

## 3.1 Calculations

Using the same basis states and circuit from part 2, we modify the circuit and add a X (bit-flip) gate to qubit 1 (the second qubit) before the CNOT. This means we are now starting with an initial state of $|00\rangle \xrightarrow{I\otimes X} |01\rangle$, then apply $H \otimes I$ then CNOT to create a Modified Bell State.

$$(H \otimes I)\,|01\rangle = H\,|0\rangle \otimes I\,|1\rangle = |+\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right), \tag{5}$$

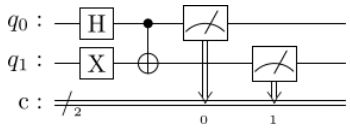
where the result of Eq. 1 was used to calculate $H\,|0\rangle$ and the main property of the Identity matrix was used to calculate $I\,|1\rangle$. As the modification done to the circuit only effects the second qubit, this means the result should be simmilar to Part 2, with

$$\text{CNOT}\,(H \otimes I)\,|01\rangle = \text{CNOT}\frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\right)\left(|01\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \tag{6}$$

As expected, this result is similar to Part 2, but now we should have a 50/50 chance of our result being in $|01\rangle$ or $|10\rangle$ instead of $|00\rangle$ or $|11\rangle$.

## 3.2 Code

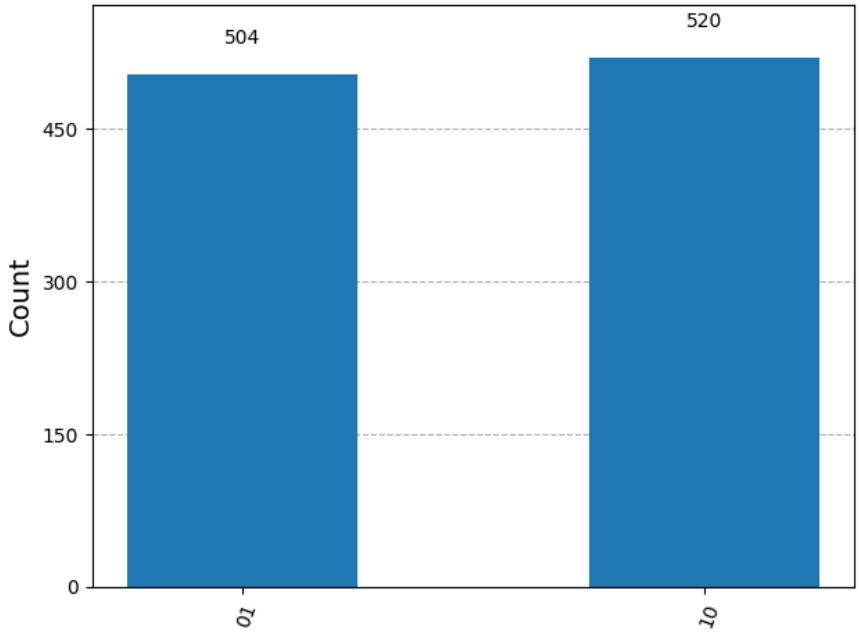
The circuit to implement the modified bell states is shown below.




For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_mod = QuantumCircuit(2,2)
qc_mod.x(1)
qc_mod.h(0)
qc_mod.cx(0,1)
qc_mod.measure(0,0)
qc_mod.measure(1,1)
shots_mod = 1024
job_mod = simulator.run([qc_mod], shots = shots_mod)
plot_histogram(job_mod.result().get_counts(0))
```

### 3.3 Results




The result of the simulation shows that the number of shots that were measured in the $|01\rangle$ state was 504 compared to the 520 shots measured in $|10\rangle$. The total number of shots ran was 1024, which means we had $\frac{504}{1024} \approx 49\%$ of the measurements result in $|01\rangle$ and $\frac{520}{1024} \approx 51\%$ of the measurements result in $|10\rangle$ which is within the 5% error margin expected. These percentages add up to 100%, which is expected as the only two results shown in the histogram are $|01\rangle$ and $|10\rangle$.

## 4 Design Challenge

### 4.1 Calculations

Using the same basis states, we need to make a modified version of the Bell State where instead of a 50/50 split between $|00\rangle$ and $|11\rangle$ we have a 75/25 split. The Hadamard gate preforms a rotaton on the qubit gate 90 degress about the y-axis. So instead we can preform a rotation about the y-axis that gives us the wanted 75/25 split. So the fllowing circuit

$$RY_\theta |0\rangle = \alpha |0\rangle + \beta |1\rangle \tag{7}$$

where we want $|\alpha|^2 = .75$ and $|\beta|^2 = .25$. Since we are rotating around the y-axs which is where we get imaginary values in the probablity ampltudes, $\alpha$ and $\beta$ should be real. From here we can say we want $\alpha = \sqrt{.75} = \frac{\sqrt{3}}{2}$ and $\beta = \sqrt{.25} = \frac{1}{2}$. Knowing that these are the values associated with the angle $\theta = \frac{\pi}{3} = 60°$, we can say,

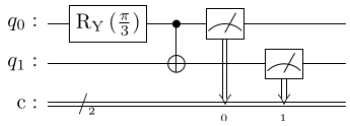
$$RY_{\frac{\pi}{3}} |0\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle . \tag{8}$$

Which means

$$\text{CNOT} \left(RY_{\frac{\pi}{3}} \otimes I\right) |00\rangle = \text{CNOT} \left( \frac{\sqrt{3}}{2} |00\rangle + \frac{1}{2} |10\rangle \right) = \frac{\sqrt{3}}{2} |00\rangle + \frac{1}{2} |11\rangle$$

### 4.2 Code

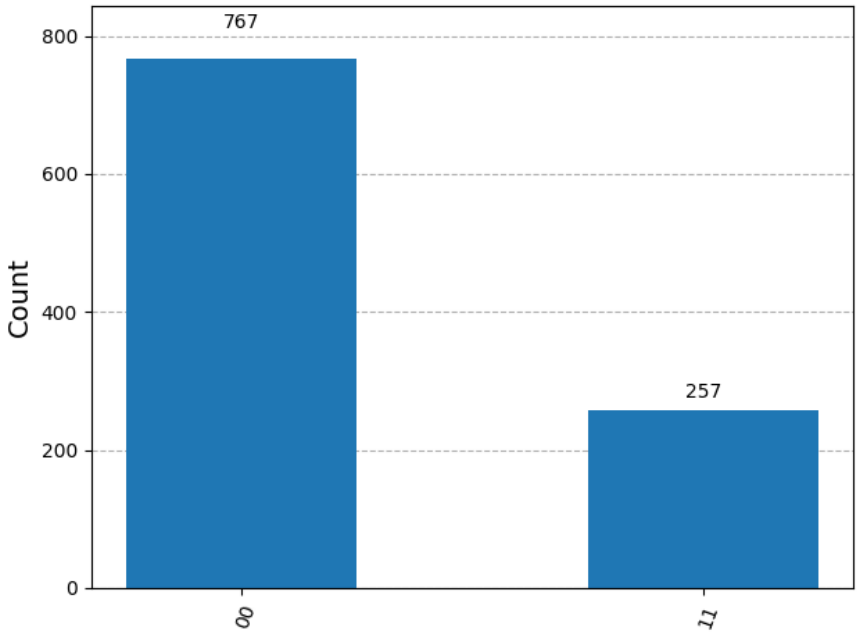
The circuit to implement the modified bell states is shown below.




For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_design = QuantumCircuit(2,2)
qc_design.ry(np.deg2rad(60),0)
qc_design.cx(0,1)
qc_design.measure(0,0)
qc_design.measure(1,1)
shots_design = 1024
job_design = simulator.run([qc_design], shots = shots_design)
plot_histogram(job_design.result().get_counts(0))
```

## 4.3 Results




The result of the simulation shows that the number of shots that were measured in the $|00\rangle$ state was 767 compared to the 257 shots measured in $|10\rangle$. The total number of shots ran was 1024, which means we had $\frac{767}{1024} \approx 74.9\%$ of the measurements result in $|00\rangle$ and $\frac{257}{1024} \approx 25.1\%$ of the measurements result in $|11\rangle$ which is within the 5% error margin expected. These percentages add up to 100%, which is expected as the only two results shown in the histogram are $|00\rangle$ and $|11\rangle$.